

House Price Prediction using Machine Learning

Aim

To implement a **House Price Prediction** model using **Linear Regression** in Python and predict house prices based on features such as area, number of bedrooms, and age of the house.

Apparatus Required

- Google Colab / Jupyter Notebook
- Python 3
- Libraries:
 - NumPy
 - Pandas
 - Scikit-learn (sklearn)

Theory

House Price Prediction is a **supervised machine learning regression** problem. The objective is to predict the price of a house based on its features. **Linear Regression** is one of the most commonly used algorithms for regression problems as it models the relationship between independent variables and the target variable.

Advantages:

- Simple and easy to implement.
- Fast and efficient.
- Suitable for predicting continuous values.

Disadvantages:

- Assumes a linear relationship between variables.
- Sensitive to outliers.

Algorithm

1. Import the required libraries.
2. Create a sample house dataset.
3. Split the dataset into training and testing sets.
4. Train the Linear Regression model.
5. Predict the house prices.
6. Calculate the Mean Squared Error (MSE).
7. Calculate the R^2 Score.
8. Display the predicted prices.

Code (Google Colab)

```
import pandas as pd

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score

# Sample House Dataset
data = {
    'Area': [1000, 1200, 1500, 1800, 2000, 2200, 2500, 3000],
    'Bedrooms': [2, 2, 3, 3, 4, 4, 4, 5],
    'Age': [20, 15, 10, 8, 5, 4, 3, 2],
    'Price': [2500000, 3000000, 4000000, 5000000,
              6000000, 6500000, 7000000, 8500000]
}

df = pd.DataFrame(data)

# Features and Target
```

```
X = df[['Area', 'Bedrooms', 'Age']]
y = df['Price']

# Split the dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.25, random_state=42
)

# Train the model
model = LinearRegression()
model.fit(X_train, y_train)

# Predict
y_pred = model.predict(X_test)

# Display Results
print("Actual House Prices:")
print(y_test.values)

print("\nPredicted House Prices:")
print(y_pred)

print("\nMean Squared Error (MSE):")
print(mean_squared_error(y_test, y_pred))

print("\nR2 Score:")
print(r2_score(y_test, y_pred))
```

Sample Output

```
*** Actual House Prices:
[3000000 6500000]

Predicted House Prices:
[3012438.09415381 6322052.71247225]

Mean Squared Error (MSE):
15909971662.331245

R2 Score:
0.9948049072123
```

Result

The **House Price Prediction** model was successfully implemented using the **Linear Regression** algorithm in Python. The model predicted house prices based on features such as area, number of bedrooms, and age of the house. The model's performance was evaluated using the **Mean Squared Error (MSE)** and **R² Score**.